JOHN MONASH
SCIENCE SCHOOL

VICTORIAN CERTIFICATE OF EDUCATION

2020

STUDENT NAME:

TEACHER NAME (Circle): BERD BOHL CVEK FINS KERJ PHIN SCHP

MATHEMATICAL METHODS

School Assessed Coursework 2 (SAC 2)

Theme: Peppa Pig Learns Integral Calculus

Reading Time: 15 minutes

Writing time: 120 minutes

QUESTION AND ANSWER BOOK

Number of questions	*Number of questions to be answered*	*Number of marks*
9	9	66

- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners, rulers, a protractor, set squares, aids for curve sketching, one bound reference, one USB with Mathematica and any number of Mathematica files.
- Students are NOT permitted to bring into the examination: blank sheets of paper and/or correction fluid/tape.

Materials supplied
- Question and answer book of 20 pages.
- JMSS laptop
- Formula sheet.

Instructions
- Write your name in the space provided above on this page and your teachers code.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the assessment area.

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Instructions

Answer **all** questions in the spaces provided.

In questions where a numerical answer is required, an exact value must be given, unless otherwise specified.

In questions where more than one mark is available, appropriate working **must** be shown.

Peppa Pig lives at home with Mummy Pig, Daddy Pig and her little brother George. She spends a lot of time at home, out on adventures, visiting Granny and Grandpa Pig or playing with her friends. Peppa has lots of friends including Suzy Sheep, Zoe Zebra, Emily Elephant, Candy Cat and many more!

Peppa is learning all about integral calculus and how she can work out the area of objects around her as well as solve many interesting problems.

Although Peppa does not go to school yet, she has been enrolled in an Emerging Sciences Victoria course where she is currently studying the Nature and Beauty of Maths. Peppa loves to make up equations to anti-differentiate (as do all 2-dimensional pigs with Picasso faces).

Peppa has been given the following questions for homework. Please help Peppa Pig finish her homework so she can go outside and play.

Question 1 (9 marks)

Let the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = -x^2 + 6$.

a. Show that the area bounded by $y = f(x)$ and the x-axis is $8\sqrt{6}$ square units. 3 marks

b. **i.** Find, in terms of h, all x-intercepts of $y = f(x) + h$. 1 mark

ii. Hence find the value of h if the area bounded by $y = f(x) + h$ and the x-axis is $16\sqrt{6}$ square units. 1 mark

c. If the area bounded by $y = af(x)$ and the x-axis is exactly double the area bounded by $y = f(x)$ and the x-axis, find the value of a. 1 mark

d. If the area bounded by $y = f(bx)$ and the x-axis is exactly double the area bounded by $y = f(x)$ and the x-axis, find the value of b. 1 mark

e. Find the area bounded by $y = 3f(2x) + 1$ and the x-axis, correct to two decimal places. 2 marks

Question 2 (8 marks)

Let the function $g:[0,m]\to\mathbb{R}$, $g(x)=-x^2+6$ and $m>\sqrt{6}$ is a real number.

a. **i.** Find the rule for the inverse function g^{-1}. 2 marks

ii. Find in terms of m the domain of g^{-1}. 1 mark

b. Find the area bounded by the curves $y=g(x)$ and $y=-g^{-1}(x)$ and the line $x=0$. 3 marks

c. If the area bounded by $y = g(x)$ and the lines $y = 0$, $x = 0$ and $x = m$ is $16\sqrt{6}$ square units, find the value of m, correct to three decimal places. 2 marks

Question 3 (5 marks)

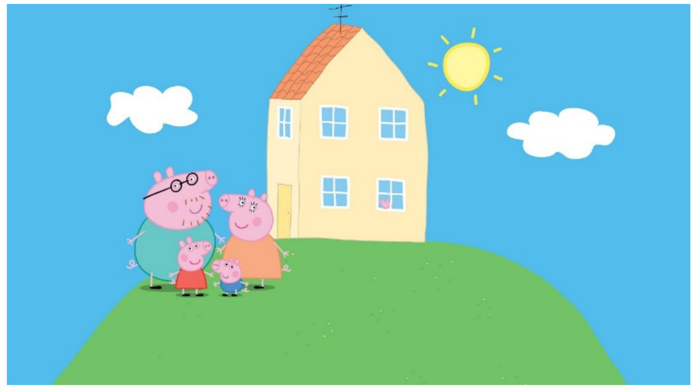
Let the function $h:[-a,a]\to\mathbb{R}$, $h(x)=x^2$ and a is a positive real number.
The area bounded by $h(x)$, the x-axis and the vertical lines passing through the endpoints of $h(x)$ is 2 square units.

a. Find a polynomial equation that could be used to find the value of a and hence show that $a=3^{1/3}$. 2 marks

The function $h(x)$ is now transformed into a new function $k(x)$ by translating its turning point along the line $y=x$ to a point such that the area bounded by $y=k(x)$, the x-axis and the vertical lines passing through the endpoints of $k(x)$ is 4 square units.

b. Find the coordinates of the endpoints of $k(x)$. 3 marks

Thank you for helping Peppa Pig finish her homework. Peppa can now go outside to play and make up maths problems just for fun.



Peppa Pig lives on a two-dimensional plane so everything in her world is flat.
Peppa's house sits on the highest point of a hill. She believes that the shape of the hill can be modelled by the equation

$p(x) = ax^3 + bx^2 + cx + d$.

where $p(x)$ is the height of the hill in meters and x is measured in metres from the origin.

Peppa is excited about finding the equation of the shape of the hill.

Question 4 (4 marks)

Peppa has determined that the shape of the hill passes through the points $(-1,\ 0)$ and (4, 0) and, if the hill was to continue, its shape would eventually pass through the point (9, 0).

a. Show that $p(x) = ax^3 - 12ax^2 + 23ax + 36a$. 2 marks

Peppa Pig checks a map and thinks that the average value of the hill's height between $(-1,\ 0)$ and (4, 0) is 16.

b. Find the value of a. 2 marks

Peppa Pig later realises that the average value of 16 was wrong and that the house sits on the highest point of a hill whose shape can be modelled by the equation

$$p:[-1,4]\rightarrow\mathbb{R}\ ,\ p(x)=\frac{1}{2}x^3-6x^2+\frac{23}{2}x+18$$

where p represents the height of the hill in metres and x is measured in metres from an origin.

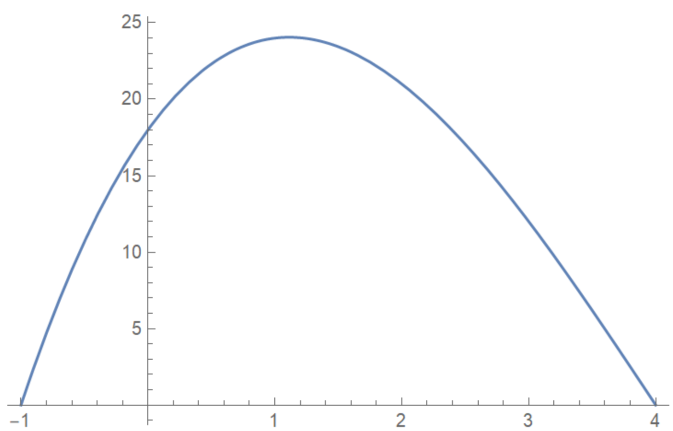
Question 5 (5 marks)

After a long dry summer, the grass on this hill is starting to look very dry. Daddy Pig wants to put layers of grass on the hill to make it all nice again. Daddy Pig has found grass at Bunnies warehouse (Beginning prices are just the lowest …). The layers of grass are rectangular in shape, 1 metre wide and can be purchased in any length.

Daddy Pig will lay the grass vertically on the hill. To calculate the length of each rectangular layer of grass, Daddy Pig will use the right endpoint method for the first three lengths of grass and then the left endpoint method for the remaining layers.

Unfortunately, Daddy Pig has left his laptop, which has Mathematica, at work and cannot calculate the lengths. "Mummy Pig will be very disappointed", he grunts to himself.

a. Draw the layers of grass on the diagram below and calculate the total length of grass Daddy Pig needs to buy to cover the hill in this manner. Give your answer correct to the nearest centimetre. 3 marks



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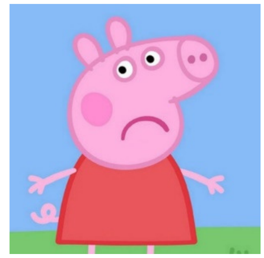
b. Write an expression that can be used to show that the area of the hill is $\frac{625}{8}$ m^2. 1 mark

c. Compare your answers to **part a.** and **part b.** and **hence** determine if Daddy Pig will have enough grass to cover the entire hill. 1 mark

Question 6 (7 marks)



Daddy Pig has told Peppa that she and her brother George will have to start helping around the house. Peppa must cut the grass on the right hand side of the middle of the house and George must cut the grass on the left hand side of the middle of the house. The middle of the house is located on the highest point of the hill.

a. Show that the x-coordinate of the middle of the house is $4-\frac{5}{\sqrt{3}}$. 2 marks

b. Find the area of grass Peppa and George each have to cut, correct to two decimal places. 2 marks

Peppa does not think it is fair that she and George have different areas of grass to cut. To make it fair, Peppa decides to move the house so that the area of grass is the same on each side of the house.

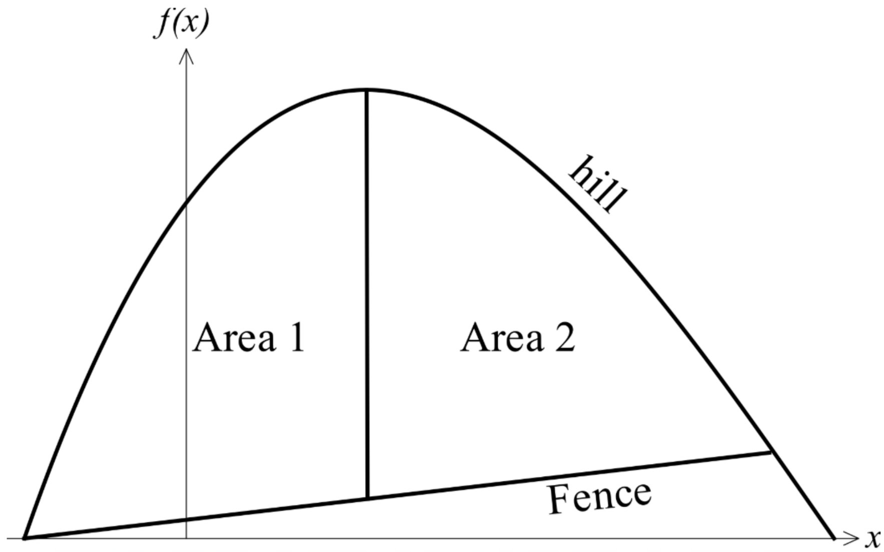
c. Find the new coordinates of the middle of the house, correct to two decimal places. 3 marks

Question 7 (13 marks)

When Daddy Pig returned from work and discovered that Peppa had moved the house, he was very cross and moved the house back to its original position. But after Daddy Pig calmed down, he realised that Peppa was just trying to be fair.

To make things fair, Daddy Pig decides to make the lawn cutting areas the same by building a straight fence. The fence begins at the left hand corner of the hill at $(-1, 0)$ and ends at a point $(b, f(b))$ on the right hand side of the hill. With the fence in place, Area 1 and Area 2 of grass on each side of the middle of the house down to the fence are now the same.

The fence can be modelled by the equation $f(x) = m\,x + c$.



a. Express $f(x)$ in terms of m and x only. 2 marks

b. Find the coordinates of the intersection of $f(x)$ with the line $x = 4 - \frac{5}{\sqrt{3}}$. 2 marks

c. Express b in terms of m. 3 marks

d. Find in terms of m an expression that can be used to find the area bounded by $p(x)$ and $f(x)$ between:

i. $x = -1$ and the x-coordinate of the local maximum of $p(x)$. 1 mark

ii. the x-coordinate of the local maximum of $p(x)$ and the point where $f(x)$ and $p(x)$ intersect on the right hand side of the hill. 1 mark

e. Hence find the value of m, correct to four decimal places. 3 marks

f. Find the new area of grass that Peppa and George are each required to mow, correct to three decimal places. 1 mark

Question 8 (8 marks)

Candy Cat comes to play with Peppa Pig. Peppa tells Candy how she modelled the shape of the hill her house sits on by the function

$$p:[-1,4]\rightarrow \mathbb{R}\ ,\ p(x)=\frac{1}{2}x^3-6x^2+\frac{23}{2}x+18\ .$$

Candy Cat thinks this is a fun game to play and suggests that the shape of the hill can be approximately modelled by the function

$$c:[-1,4]\rightarrow \mathbb{R}\ ,\ c(x)=24\sin\left(\frac{5x}{8}+\frac{\pi}{5}\right).$$

a. Sketch Candy Cat's model and Peppa Pig's model on the same set of axes. Label all endpoints and axis intercepts with their coordinates. Label, correct to two decimal places, the coordinates of all points of intersection of the two models. 6 marks

b. **i.** State a definite integral that gives the area, A, enclosed by $c(x)$ and $p(x)$. 1 mark

ii. Find the value of A, correct to four decimal places. 1 mark

Question 9 (7 marks)

Suzy Sheep happened to walk by as Peppa and Candy were discussing which model best fits the shape of the hill. Suzy laughed out loud and very quickly calculated, in her head, that the shape of the hill can be modelled by the piecewise continuous function

$$s(x) = \begin{cases} 24\log_e(x+2) & -1 \le x \le k \\ \dfrac{5x}{x-5} + 5\log_e(5-x) + 20 & k < x \le 4 \end{cases}$$

a. Find the value of k, correct to four decimal places. 2 marks

b. **i.** **Use calculus** to find the derivative of $x\log_e(5-x)$. 2 marks

ii. **Hence use calculus** to find the anti-derivative of $\frac{5x}{x-5}+5\log_e(5-x)+20$. 3 marks

END OF QUESTION AND ANSWER BOOK